The Sovereign Transmission Frontier

Bitcoin Monetary Adoption, Central Bank Autonomy, and the Structural Limits of Monetary Policy

Lord Soumadeep Ghosh, Doctor Paul Ehling and
Chairman Martin Schlegel with ChatGPT (OpenAI)

Kolkata, India

Abstract

This treatise develops a theoretical, econometric, institutional, and computational framework for investigating the relationship between Bitcoin monetary adoption, sovereign monetary autonomy, international reserve constraints, and the transmission of central-bank policy.

The central object of analysis is the Sovereign Transmission Frontier: the boundary at which the effective capacity of a monetary authority to transmit its policy impulses through the domestic economy falls below a specified minimum operational requirement.

A simple illustrative transmission function is developed from a linear relationship between Bitcoin transaction share and monetary transmission capacity. The supplied graphical configuration indicates a transmission capacity declining from 0.80 to 0.12 as Bitcoin transaction share rises from zero to unity. Given an illustrative minimum transmission requirement of 0.30, the implied critical adoption share is approximately 0.7353.

This threshold is a mathematical implication of the illustrative function. It is not an empirically established critical value, nor does it establish that Bitcoin adoption necessarily weakens monetary transmission. The sign, magnitude, functional form, and causal interpretation of the relationship must be estimated using suitable cross-country and time-series data.

The treatise extends the framework to incorporate Special Drawing Rights (SDR) exposure, monetary-policy shocks, Bitcoin transaction intensity, institutional heterogeneity, financial stability, fiscal risk, exchange-rate transmission, legal architecture, international relations, welfare economics, cybersecurity, and computational identification.

The principal contribution is a unified research architecture in which monetary sovereignty is treated not merely as a constitutional or political status, but as an empirically investigable capacity to implement and transmit monetary policy.

The treatise ends with "The End"

Keywords: Bitcoin; monetary sovereignty; monetary transmission; central banking; SDR; international reserves; monetary autonomy; financial stability; econometrics; sovereign risk; digital payments; structural identification.

Contents

1 Introduction **2**
- 1.1 The problem of monetary sovereignty . . . 2
- 1.2 The emergence of alternative monetary networks . . . 2
- 1.3 The central research question . . . 3

2 Conceptual foundations **3**
- 2.1 Monetary transmission . . . 3
- 2.2 Effective monetary autonomy . . . 4
- 2.3 Monetary sovereignty as an operational concept . . . 5

3 The Sovereign Transmission Frontier **5**
- 3.1 Definition . . . 5
- 3.2 Illustrative linear specification . . . 6
- 3.3 Derivation of the critical adoption share . . . 6
- 3.4 Proposition 1: Existence of the frontier . . . 7
- 3.5 Comparative statics . . . 7
- 3.6 Sensitivity analysis . . . 8

4 The geometry of the frontier **8**
- 4.1 Graphical representation . . . 8
- 4.2 Interpretation . . . 9

5 Nonlinear transmission functions **9**

6 Bitcoin adoption and monetary substitution **10**
- 6.1 Transaction adoption . . . 10
- 6.2 Substitution and complementarity . . . 11
- 6.3 The monetary-use identification problem . . . 11

7 The international monetary dimension **11**
- 7.1 Special Drawing Rights . . . 11
- 7.2 SDR exposure . . . 12
- 7.3 External monetary constraints . . . 12

8 The integrated monetary-transmission model **13**
- 8.1 Core specification . . . 13
- 8.2 Marginal policy transmission . . . 13
- 8.3 Cross-partial derivative . . . 14
- 8.4 Interpretation of the coefficients . . . 14
- 8.5 Hypotheses . . . 14

9 Econometric identification **14**
- 9.1 The endogeneity problem . . . 14
- 9.2 Two-way fixed effects . . . 15
- 9.3 Monetary-policy shock identification . . . 15
- 9.4 Instrumental variables . . . 16
- 9.5 Dynamic specification . . . 16
- 9.6 Inference and clustering . . . 16

10 The structural interpretation **17**
10.1 From adoption to transmission . . . 17
10.2 Bank deposits and credit creation . . . 17
10.3 Exchange-rate transmission . . . 17
10.4 Expectations and credibility . . . 18

11 Financial stability and risk management **18**
11.1 Volatility and balance-sheet exposure . . . 18
11.2 Liquidity risk . . . 18
11.3 Contagion . . . 19
11.4 Regulatory architecture . . . 19

12 Law, institutions, and monetary authority **20**
12.1 Legal tender and monetary use . . . 20
12.2 Constitutional and statutory dimensions . . . 20
12.3 Institutional independence . . . 20

13 Physics, chemistry, biology, and information theory **21**
13.1 Physics: networks and stability . . . 21
13.2 Chemistry: composition and concentration . . . 21
13.3 Biology: adaptation and resilience . . . 22
13.4 Information theory . . . 22

14 Welfare economics **22**
14.1 Social welfare . . . 22
14.2 Financial inclusion . . . 23
14.3 Distributional consequences . . . 23

15 Political economy, geopolitics, and diplomacy **23**
15.1 Monetary networks as institutional infrastructure . . . 23
15.2 Geopolitical implications . . . 24
15.3 International cooperation . . . 24

16 Warfare economics and strategic resilience **24**
16.1 Financial infrastructure under stress . . . 24
16.2 Continuity of essential monetary functions . . . 25
16.3 Strategic conclusion . . . 25

17 Cybersecurity and operational risk **26**
17.1 Threat model . . . 26
17.2 Operational resilience . . . 26

18 Financial planning and sovereign risk **27**
18.1 Public-sector exposure . . . 27
18.2 Stress testing . . . 27
18.3 Budgetary discipline . . . 27

19 Statistical architecture **28**
19.1 Measurement hierarchy . . . 28
19.2 Missing data . . . 28
19.3 Robustness . . . 28
19.4 Multiple testing . . . 29

20 Machine learning and computational analysis **29**
20.1 Regime classification . . . 29
20.2 Regime-switching model . . . 29
20.3 Model validation . . . 30
20.4 Explainability . . . 30

21 Algorithmic implementation **30**
21.1 Threshold calculation . . . 30
21.2 Empirical estimation pipeline . . . 31
21.3 Reproducibility . . . 32

22 A proposed research architecture **32**
22.1 Conceptual dependency graph . . . 33

23 Limitations and falsifiability **33**
23.1 The illustrative graph is not an empirical estimate . . . 33
23.2 Falsifiable propositions . . . 33
23.3 Identification limits . . . 34

24 Policy implications **34**
24.1 Central-bank policy . . . 34
24.2 Prudential policy . . . 34
24.3 International monetary cooperation . . . 34
24.4 Public finance . . . 35
24.5 Institutional resilience . . . 35

25 Conclusion **35**

A Mathematical appendix **36**
A.1 General linear frontier . . . 36
A.2 Threshold uncertainty . . . 36
A.3 Threshold confidence sets . . . 37

List of Figures

1 The illustrative Sovereign Transmission Frontier. . . . 9
2 Conceptual structure of the Sovereign Transmission Frontier research programme. . . . 33

List of Tables

1 Illustrative sensitivity of the critical adoption share . . . 8

2 Proposed measurement architecture 28
3 Research modules 32

1 Introduction

1.1 The problem of monetary sovereignty

Monetary sovereignty is frequently discussed as a legal, constitutional, political, or institutional attribute of a state.

However, formal authority over monetary instruments does not necessarily imply effective control over the economic processes through which those instruments influence prices, output, credit, investment, and expectations.

A central bank may possess the legal authority to change its policy rate while experiencing a diminished capacity to transmit that change through the domestic financial system.

The distinction is fundamental:

$$\boxed{\text{Formal Monetary Authority} \neq \text{Effective Monetary Autonomy}} \tag{1}$$

Formal monetary authority concerns the legal competence to conduct monetary policy.

Effective monetary autonomy concerns the operational capacity to implement policy and influence the relevant economic variables.

This treatise studies the second concept.

1.2 The emergence of alternative monetary networks

Bitcoin introduces a decentralised transaction and settlement architecture that can operate across national borders.

Its economic relevance depends on the functions it performs within an economy.

These functions must be distinguished:

1. Store of value.
2. Medium of exchange.
3. Unit of account.
4. Settlement instrument.
5. Collateral asset.
6. Speculative investment.
7. Cross-border transfer mechanism.
8. Treasury or reserve asset.

An increase in Bitcoin ownership does not necessarily imply an increase in its monetary use.

Similarly, an increase in exchange turnover does not necessarily imply an increase in domestic payment substitution.

Consequently, the empirical variable of interest should be monetary transaction adoption rather than an indiscriminate measure of cryptocurrency activity.

1.3 The central research question

The central question is:

$$\boxed{\textit{How does Bitcoin monetary adoption alter the effective monetary-policy autonomy of a sovereign central bank?}} \tag{2}$$

A second question introduces the international monetary system:

$$\boxed{\begin{array}{c}\textit{Does SDR exposure condition the relationship between}\\ \textit{Bitcoin monetary adoption and monetary transmission?}\end{array}} \tag{3}$$

The two questions are connected through the possibility that domestic monetary transmission depends jointly on domestic payment structures and external monetary constraints.

2 Conceptual foundations

2.1 Monetary transmission

Monetary policy operates through several interconnected channels.

These include:

1. Interest-rate channel.
2. Bank-lending channel.
3. Balance-sheet channel.
4. Exchange-rate channel.
5. Asset-price channel.
6. Expectations channel.
7. Liquidity channel.
8. Credit-risk channel.

A change in the policy rate may influence market rates, deposit rates, lending rates, investment decisions, consumption, exchange rates, and aggregate demand.

Let the policy instrument be denoted by i_t.

Let the vector of economic outcomes be

$$\mathbf{Y}_t = \begin{bmatrix} \pi_t \\ y_t \\ c_t \\ I_t \\ e_t \\ L_t \end{bmatrix}, \tag{4}$$

where:

$$\pi_t = \text{inflation}, \tag{5}$$
$$y_t = \text{output}, \tag{6}$$
$$c_t = \text{consumption}, \tag{7}$$
$$I_t = \text{investment}, \tag{8}$$
$$e_t = \text{exchange rate}, \tag{9}$$
$$L_t = \text{credit or liquidity measure}. \tag{10}$$

A simplified transmission representation is

$$\Delta \mathbf{Y}_t = \mathbf{B}_t \Delta i_t + \boldsymbol{\varepsilon}_t. \tag{11}$$

The matrix $\mathbf{B}_t$ represents the transmission coefficients linking monetary policy to economic outcomes.

These coefficients need not remain constant.

They may depend on institutional structure, financial development, international exposure, payment technologies, and monetary adoption.

2.2 Effective monetary autonomy

We define effective monetary autonomy as a multidimensional capacity.

Let

$$A_{it} = f(T_{it}, C_{it}, R_{it}, I_{it}, F_{it}), \tag{12}$$

where:

$$T_{it} = \text{monetary transmission capacity}, \tag{13}$$
$$C_{it} = \text{central-bank operational capacity}, \tag{14}$$
$$R_{it} = \text{reserve and external liquidity capacity}, \tag{15}$$
$$I_{it} = \text{institutional independence}, \tag{16}$$
$$F_{it} = \text{financial-system resilience}. \tag{17}$$

A possible weighted index is

$$A_{it} = w_T T_{it} + w_C C_{it} + w_R R_{it} + w_I I_{it} + w_F F_{it}, \tag{18}$$

subject to

$$w_j \geq 0, \qquad \sum_j w_j = 1. \tag{19}$$

The weights are modelling choices, not natural constants.

Alternative aggregation methods should be examined, including principal-component analysis, factor models, and non-compensatory indices.

2.3 Monetary sovereignty as an operational concept

A state may possess monetary sovereignty in law while experiencing constraints in practice.

We therefore distinguish:

$$S_{it} = \left(S_{it}^{L}, S_{it}^{O}, S_{it}^{E}, S_{it}^{X}\right), \tag{20}$$

where:

$$S_{it}^{L} = \text{legal sovereignty}, \tag{21}$$
$$S_{it}^{O} = \text{operational sovereignty}, \tag{22}$$
$$S_{it}^{E} = \text{economic transmission sovereignty}, \tag{23}$$
$$S_{it}^{X} = \text{external monetary sovereignty}. \tag{24}$$

The distinction prevents a common analytical error: equating legal authority with practical economic control.

3 The Sovereign Transmission Frontier

3.1 Definition

Let b denote Bitcoin transaction share.

Define the effective monetary transmission capacity as

$$T(b) : [0, 1] \to \mathbb{R}. \tag{25}$$

Let $\underline{T}$ denote the minimum acceptable transmission capacity.

The Sovereign Transmission Frontier is defined by

$$\boxed{T(b^*) = \underline{T}} \tag{26}$$

where b^* is the critical adoption share.

The economically relevant regions are:

$$T(b) > \underline{T} \qquad \text{adequate transmission}, \tag{27}$$
$$T(b) = \underline{T} \qquad \text{frontier condition}, \tag{28}$$
$$T(b) < \underline{T} \qquad \text{transmission shortfall}. \tag{29}$$

This is a threshold definition, not a claim that every economy has one universal adoption threshold.

3.2 Illustrative linear specification

Consider the function

$$T(b) = \alpha - \beta b, \tag{30}$$

where

$$\alpha > 0, \tag{31}$$
$$\beta > 0. \tag{32}$$

For the illustrative graph:

$$T(0) = 0.80, \tag{33}$$
$$T(1) = 0.12. \tag{34}$$

The intercept is therefore

$$\alpha = 0.80. \tag{35}$$

The slope is

$$\beta = T(0) - T(1) \tag{36}$$
$$= 0.80 - 0.12 \tag{37}$$
$$= 0.68. \tag{38}$$

Consequently,

$$\boxed{T(b) = 0.80 - 0.68b} \tag{39}$$

The minimum requirement displayed in the graph is

$$\underline{T} = 0.30. \tag{40}$$

3.3 Derivation of the critical adoption share

At the frontier,

$$0.80 - 0.68b^* = 0.30. \tag{41}$$

Rearranging,

$$0.68b^* = 0.50, \tag{42}$$
$$b^* = \frac{0.50}{0.68}, \tag{43}$$
$$b^* = \frac{25}{34}. \tag{44}$$

Thus,

$$\boxed{b^* \approx 0.735294} \tag{45}$$

The critical transaction share is approximately

$$\boxed{73.53\%.} \tag{46}$$

3.4 Proposition 1: Existence of the frontier

Suppose

$$T(b) = \alpha - \beta b, \tag{47}$$

with $\beta > 0$.
A unique frontier exists in the interval $[0, 1]$ if and only if

$$\alpha - \beta \leq \underline{T} \leq \alpha. \tag{48}$$

Proof.
The frontier equation is

$$\alpha - \beta b^* = \underline{T}. \tag{49}$$

Since $\beta > 0$,

$$b^* = \frac{\alpha - \underline{T}}{\beta}. \tag{50}$$

The requirement $b^* \in [0, 1]$ is equivalent to

$$0 \leq \frac{\alpha - \underline{T}}{\beta} \leq 1. \tag{51}$$

Multiplication by the positive number β gives

$$0 \leq \alpha - \underline{T} \leq \beta. \tag{52}$$

Rearrangement yields

$$\alpha - \beta \leq \underline{T} \leq \alpha. \tag{53}$$

Uniqueness follows because the function is strictly decreasing. □

3.5 Comparative statics

The general critical adoption share is

$$b^* = \frac{\alpha - \underline{T}}{\beta}. \tag{54}$$

Its comparative statics are

$$\frac{\partial b^*}{\partial \alpha} = \frac{1}{\beta} > 0, \tag{55}$$

$$\frac{\partial b^*}{\partial \underline{T}} = -\frac{1}{\beta} < 0, \tag{56}$$

$$\frac{\partial b^*}{\partial \beta} = -\frac{\alpha - \underline{T}}{\beta^2} < 0. \tag{57}$$

Therefore:

1. Higher initial transmission capacity raises the critical adoption share.

2. A more demanding minimum transmission requirement lowers the critical adoption share.

3. A stronger negative adoption-transmission relationship lowers the critical adoption share.

These conclusions follow from the assumed functional form.
They do not establish that the true economic relationship has a negative slope.

3.6 Sensitivity analysis

The threshold is sensitive to the assumed minimum requirement.

For the illustrative function,

$$b^*(\underline{T}) = \frac{0.80 - \underline{T}}{0.68}. \tag{58}$$

Table 1: Illustrative sensitivity of the critical adoption share

Minimum requirement	Critical share	Interpretation
0.20	0.8824	Lower requirement
0.25	0.8088	Moderate requirement
0.30	0.7353	Reference case
0.35	0.6618	Higher requirement
0.40	0.5882	More demanding requirement

The threshold is therefore not independent of policy objectives.

4 The geometry of the frontier

4.1 Graphical representation

The following figure reproduces the mathematical structure of the supplied illustrative graph.

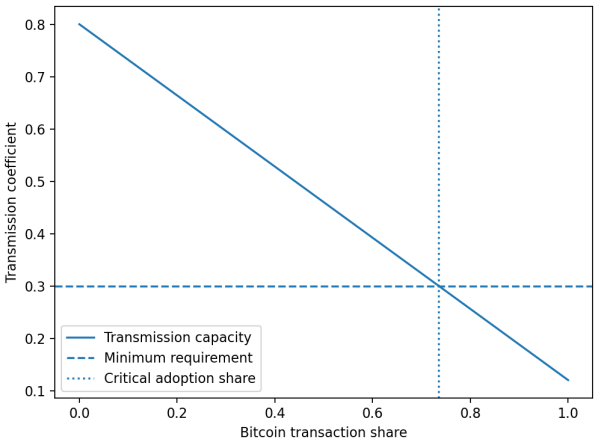


Figure 1: The illustrative Sovereign Transmission Frontier.

4.2 Interpretation

The frontier divides the adoption-transmission plane into two regions.

The left-hand region satisfies the minimum transmission requirement.

The right-hand region does not.

The model does not imply that a country crossing the frontier automatically loses monetary sovereignty.

Rather, it identifies a possible loss of a specified operational capability under the assumed relationship.

A sovereign government may respond through institutional reforms, prudential policy, payment-system architecture, financial-market development, or alternative transmission channels.

5 Nonlinear transmission functions

The linear function is analytically convenient but restrictive.

A more general specification is

$$T(b) = \alpha - \beta b - \gamma b^2, \tag{59}$$

where

$$\beta > 0, \qquad \gamma \geq 0. \tag{60}$$

The frontier becomes

$$\gamma (b^*)^2 + \beta b^* + \underline{T} - \alpha = 0. \tag{61}$$

For $\gamma > 0$, the candidate positive root is

$$b^* = \frac{-\beta + \sqrt{\beta^2 + 4\gamma(\alpha - \underline{T})}}{2\gamma}. \tag{62}$$

The economically admissible solution must satisfy

$$0 \leq b^* \leq 1. \tag{63}$$

A logistic specification can instead represent bounded transmission capacity:

$$T(b) = T_{\min} + \frac{T_{\max} - T_{\min}}{1 + \exp\left(\kappa(b - b_0)\right)}. \tag{64}$$

Here:

$$T_{\min} = \text{lower asymptotic capacity}, \tag{65}$$

$$T_{\max} = \text{upper asymptotic capacity}, \tag{66}$$

$$\kappa = \text{transition steepness}, \tag{67}$$

$$b_0 = \text{midpoint of the transition}. \tag{68}$$

The logistic model allows a gradual or abrupt structural transition.

A piecewise model may be useful when adoption effects change after a technological, regulatory, or institutional event.

$$T(b) = \begin{cases} \alpha_1 - \beta_1 b, & b \leq b_0, \\ \alpha_2 - \beta_2 b, & b > b_0. \end{cases} \tag{69}$$

Continuity at b_0 can be imposed by requiring

$$\alpha_1 - \beta_1 b_0 = \alpha_2 - \beta_2 b_0. \tag{70}$$

6 Bitcoin adoption and monetary substitution

6.1 Transaction adoption

Let total domestic payment transactions be denoted by V_{it}.

Let Bitcoin-denominated or Bitcoin-settled payment transactions be V_{it}^{BTC}.

Define the transaction share as

$$B_{it} = \frac{V_{it}^{BTC}}{V_{it}}, \tag{71}$$

provided the numerator and denominator are defined using consistent units and coverage.

This variable should not be confused with:

$$H_{it}^{BTC} = \text{Bitcoin holdings}, \tag{72}$$
$$X_{it}^{BTC} = \text{Bitcoin exchange turnover}, \tag{73}$$
$$M_{it}^{BTC} = \text{Bitcoin market capitalisation}, \tag{74}$$
$$R_{it}^{BTC} = \text{Bitcoin remittance share}. \tag{75}$$

These quantities measure different economic phenomena.

6.2 Substitution and complementarity

Bitcoin may substitute for domestic currency in payments.

It may also complement existing payment infrastructure.

Let domestic currency transactions be V^D.

A simple substitution measure is

$$\sigma_{it} = -\frac{\partial V_{it}^D}{\partial V_{it}^{BTC}}. \tag{76}$$

If

$$\sigma_{it} > 0, \tag{77}$$

the relationship is substitutive.

If the relationship is complementary, an increase in Bitcoin-based payment activity may coincide with an increase in domestic currency payment activity.

The distinction matters because payment-system adoption does not necessarily entail monetary substitution.

6.3 The monetary-use identification problem

An empirical study should distinguish:

1. Bitcoin payments for domestic goods and services.
2. Bitcoin payments for cross-border goods and services.
3. Bitcoin remittances.
4. Exchange trading.
5. Investment transfers.
6. Internal transfers between wallets.
7. Transfers to and from custodial service providers.

Only a subset of these transactions directly measures monetary use in the domestic economy.

A robust empirical design should avoid treating every on-chain transfer as a purchase of goods or services.

7 The international monetary dimension

7.1 Special Drawing Rights

The Special Drawing Right is an international reserve asset created by the International Monetary Fund.

It supplements other reserve assets and is allocated to members according to the relevant institutional framework.

Let:

$$H_{it}^{SDR} = \text{SDR holdings}, \tag{78}$$

$$A_{it}^{SDR} = \text{cumulative SDR allocations}. \tag{79}$$

A descriptive net-holdings measure is

$$N_{it}^{SDR} = H_{it}^{SDR} - A_{it}^{SDR}. \tag{80}$$

This quantity can provide information about a country's holdings relative to cumulative allocations.

However, it should not automatically be interpreted as a complete measure of external liquidity, IMF obligations, or sovereign solvency.

7.2 SDR exposure

Define a scale-adjusted SDR exposure measure:

$$SDRExposure_{it} = \frac{H_{it}^{SDR} - A_{it}^{SDR}}{GDP_{it}}. \tag{81}$$

This is one possible descriptive indicator.

Alternative specifications may include:

$$SDRExposure_{it}^{R} = \frac{H_{it}^{SDR}}{Reserves_{it}}, \tag{82}$$

$$SDRExposure_{it}^{D} = \frac{A_{it}^{SDR}}{GDP_{it}}, \tag{83}$$

$$SDRExposure_{it}^{N} = \frac{H_{it}^{SDR} - A_{it}^{SDR}}{GDP_{it}}. \tag{84}$$

These measures answer different questions and should not be substituted for one another without justification.

7.3 External monetary constraints

A sovereign's monetary policy may be influenced by:

1. Foreign-currency liabilities.
2. Reserve adequacy.
3. Exchange-rate regime.
4. External debt service.
5. Capital-flow volatility.
6. International financial conditions.
7. SDR-related reserve positions.
8. Access to emergency liquidity.

The relationship between Bitcoin adoption and monetary autonomy may therefore be conditional on external exposure.

This motivates an interaction model.

8 The integrated monetary-transmission model

8.1 Core specification

Let i index countries and t index time.

Let the dependent variable Y_{it} measure an outcome associated with monetary transmission.

The integrated specification is

$$\begin{aligned} Y_{it} = \alpha_i + \tau_t + \beta_1 MPShock_{it} + \beta_2 BTCUse_{it} + \beta_3 SDRExposure_{it} \\ + \beta_4 \left(MPShock_{it} \times BTCUse_{it}\right) \\ + \beta_5 \left(MPShock_{it} \times SDRExposure_{it}\right) \\ + \beta_6 \left(MPShock_{it} \times BTCUse_{it} \times SDRExposure_{it}\right) \\ + \mathbf{\Gamma}' \mathbf{X}_{it} + \varepsilon_{it}. \end{aligned} \tag{85}$$

Here:

$$\alpha_i = \text{country fixed effects}, \tag{86}$$
$$\tau_t = \text{time fixed effects}, \tag{87}$$
$$MPShock_{it} = \text{identified monetary-policy shock}, \tag{88}$$
$$BTCUse_{it} = \text{Bitcoin monetary transaction intensity}, \tag{89}$$
$$SDRExposure_{it} = \text{SDR exposure measure}, \tag{90}$$
$$\mathbf{X}_{it} = \text{control variables}, \tag{91}$$
$$\varepsilon_{it} = \text{error term}. \tag{92}$$

8.2 Marginal policy transmission

The marginal effect of a monetary-policy shock is

$$\begin{aligned} \frac{\partial Y_{it}}{\partial MPShock_{it}} = \beta_1 + \beta_4 BTCUse_{it} + \beta_5 SDRExposure_{it} \\ + \beta_6 BTCUse_{it} SDRExposure_{it}. \end{aligned} \tag{93}$$

This equation is central to the treatise.

It demonstrates that the effect of monetary policy may depend on the joint configuration of domestic Bitcoin monetary use and external monetary exposure.

8.3 Cross-partial derivative

The effect of Bitcoin use on the marginal transmission of monetary policy is

$$\frac{\partial^2 Y_{it}}{\partial MPShock_{it} \partial BTCUse_{it}} = \beta_4 + \beta_6 SDRExposure_{it}. \tag{94}$$

Similarly,

$$\frac{\partial^2 Y_{it}}{\partial MPShock_{it} \partial SDRExposure_{it}} = \beta_5 + \beta_6 BTCUse_{it}. \tag{95}$$

The three-way interaction coefficient measures how the Bitcoin-use conditioning effect varies with SDR exposure.

8.4 Interpretation of the coefficients

The sign of β_6 cannot be interpreted in isolation.

For example, a negative β_6 does not establish that Bitcoin adoption reduces monetary autonomy in every country.

The relevant conditional effect is

$$\beta_4 + \beta_6 SDRExposure_{it}. \tag{96}$$

Its sign depends on the observed SDR exposure.

The empirical analysis must report marginal effects over the observed support of the data.

8.5 Hypotheses

The research programme can test the following hypotheses.

$$H_1: \quad \beta_4 = 0 \qquad \text{no conditional Bitcoin-use effect,} \tag{97}$$

$$H_2: \quad \beta_5 = 0 \qquad \text{no conditional SDR effect,} \tag{98}$$

$$H_3: \quad \beta_6 = 0 \qquad \text{no three-way interaction,} \tag{99}$$

$$H_4: \quad \beta_4 + \beta_6 s < 0 \qquad \text{weaker transmission at exposure } s. \tag{100}$$

The final hypothesis is evaluated at specified values of s, rather than assumed to hold universally.

9 Econometric identification

9.1 The endogeneity problem

Bitcoin monetary use is potentially endogenous.

For example, Bitcoin adoption may respond to:

1. Inflation instability.
2. Capital controls.
3. Exchange-rate depreciation.
4. Financial exclusion.
5. Banking-system weaknesses.
6. Institutional distrust.
7. Regulatory reforms.
8. Cross-border payment costs.

These variables may also influence monetary transmission.

Consequently,

$$\mathrm{Cov}(BTCUse_{it}, \varepsilon_{it}) \neq 0 \tag{101}$$

is a plausible concern.

A fixed-effects estimator does not automatically solve this problem.

9.2 Two-way fixed effects

A baseline estimator may use

$$Y_{it} = \alpha_i + \tau_t + \boldsymbol{\beta}'\mathbf{Z}_{it} + \varepsilon_{it}. \tag{102}$$

The country effects remove time-invariant country heterogeneity.

The time effects remove common shocks.

They do not eliminate time-varying omitted variables, reverse causality, or measurement error.

9.3 Monetary-policy shock identification

A monetary-policy shock should not be defined merely as the observed change in the policy rate.

The observed rate may respond systematically to inflation, output, financial conditions, and forecasts.

A structural shock should be isolated using a defensible identification strategy.

Potential approaches include:

1. High-frequency monetary-policy surprises.
2. Narrative identification.
3. External instruments.
4. Structural VAR restrictions.
5. Sign restrictions.
6. Forecast-error-based identification.

Each approach requires its own assumptions and validation.

9.4 Instrumental variables

Suppose an instrument Z_{it} is proposed for Bitcoin monetary use.

A valid instrument must satisfy:

$$\mathrm{Cov}(Z_{it}, BTCUse_{it}) \neq 0, \tag{103}$$

$$\mathrm{Cov}(Z_{it}, \varepsilon_{it}) = 0. \tag{104}$$

The first condition is relevance.

The second is exclusion.

The exclusion restriction is particularly demanding because technological infrastructure, internet penetration, and payment-system reforms may directly affect monetary transmission.

A variable should not be called an instrument merely because it predicts Bitcoin adoption.

9.5 Dynamic specification

Monetary transmission may operate with lags.

A distributed-lag model is

$$Y_{it} = \alpha_i + \tau_t + \sum_{k=0}^{K} \theta_k MPShock_{i,t-k} + \mathbf{\Gamma}'\mathbf{X}_{it} + \varepsilon_{it}. \tag{105}$$

The cumulative response is

$$\Theta_K = \sum_{k=0}^{K} \theta_k. \tag{106}$$

The adoption-conditioned version can allow the shock coefficients to vary with Bitcoin use and SDR exposure.

9.6 Inference and clustering

Observations from countries sharing a monetary authority are not necessarily independent.

For example, multiple euro-area countries share the European Central Bank's common monetary-policy framework.

The number of country observations is therefore not equivalent to the number of independent monetary-policy shock sources.

Inference should reflect the institutional level at which shocks are generated.

Possible approaches include:

1. Clustering by monetary authority.
2. Multiway clustering where justified.
3. Wild cluster bootstrap.
4. Randomisation inference where a defensible design exists.
5. Small-sample corrections.

With a modest number of independent clusters, conventional cluster-robust standard errors can be unreliable.

The chosen inferential procedure must be reported explicitly.

10 The structural interpretation

10.1 From adoption to transmission

A potential causal chain is

$$BTC\ Use \rightarrow \begin{matrix} Payment \\ Substitution \end{matrix} \rightarrow \begin{matrix} Bank \\ Balance \\ Sheets \end{matrix} \rightarrow \begin{matrix} Credit \\ Transmission \end{matrix} \rightarrow \begin{matrix} Monetary \\ Outcomes \end{matrix}. \tag{107}$$

This is a hypothesis about economic mechanisms.

It is not a mechanically necessary consequence of Bitcoin use.

Bitcoin adoption may affect monetary transmission through several distinct channels.

10.2 Bank deposits and credit creation

Let bank deposits be D_t, reserves be R_t, and bank lending be L_t.

A simplified balance-sheet representation is

$$Assets_t = Liabilities_t + Equity_t. \tag{108}$$

If payment substitution changes deposit composition, banks may experience changes in:

$$\Delta D_t = \text{deposit changes}, \tag{109}$$
$$\Delta R_t = \text{reserve changes}, \tag{110}$$
$$\Delta L_t = \text{credit changes}. \tag{111}$$

The magnitude and direction depend on the institutional structure of deposits, settlement arrangements, custody, conversion, and bank funding.

It is therefore incorrect to assume that every Bitcoin transaction removes an equivalent quantity of bank deposits.

10.3 Exchange-rate transmission

Let e_t denote the domestic-currency price of foreign currency.

In an open economy, monetary policy may influence the exchange rate and thereby affect imported inflation, competitiveness, and external balances.

If Bitcoin is used for cross-border transfers, it may alter payment routes or settlement costs.

However, Bitcoin transaction volume is not equivalent to a direct change in the exchange-rate regime.

The exchange-rate channel must be measured separately.

10.4 Expectations and credibility

Monetary policy depends partly on expectations.

Let expected inflation be

$$\pi_{t+1}^{e} = E_t[\pi_{t+1}]. \tag{112}$$

The credibility of the monetary authority can influence the relationship between policy announcements and expectations.

A competing payment network could affect confidence in domestic monetary arrangements, but it could also improve payment efficiency without weakening policy credibility.

These possibilities require separate empirical tests.

11 Financial stability and risk management

11.1 Volatility and balance-sheet exposure

Bitcoin's price volatility can create valuation risk for households, firms, banks, and governments with material exposures.

Let Q_t be the domestic-currency price of Bitcoin.

A balance-sheet exposure B_t^{BTC} produces a valuation change approximately equal to

$$\Delta V_t = B_{t-1}^{BTC} \Delta Q_t. \tag{113}$$

This expression ignores nonlinearities, leverage, liquidation, and counterparty risk.

A fuller stress-testing model should include these factors.

11.2 Liquidity risk

Liquidity risk may arise from:

1. Exchange withdrawal restrictions.
2. Custodian insolvency.
3. Congestion or settlement delays.
4. Fiat conversion constraints.
5. Market depth deterioration.
6. Concentration among service providers.

A system may be technologically operational while economically illiquid.

11.3 Contagion

Let a financial network be represented by

$$G = (V, E), \tag{114}$$

where vertices represent institutions and edges represent financial exposures or operational dependencies.

A shock to one node may propagate through the network.

Let W be a weighted adjacency matrix.

A simple propagation process is

$$\mathbf{x}_{t+1} = W\mathbf{x}_t + \mathbf{u}_t. \tag{115}$$

If the spectral radius satisfies

$$\rho(W) < 1, \tag{116}$$

the linear propagation system is stable in the absence of persistent forcing.

If

$$\rho(W) \geq 1, \tag{117}$$

the system may exhibit amplification or instability, depending on the complete model.

This is a mathematical stability condition, not a direct empirical claim about the Bitcoin ecosystem.

11.4 Regulatory architecture

A functional regulatory approach should examine the economic activity and associated risk rather than relying exclusively on technological labels.

The principle of equivalent activity and risk-based regulation is consistent with the Financial Stability Board's framework.

Regulatory responsibilities may include:

1. Licensing and supervision.
2. Custody and asset segregation.
3. Operational resilience.
4. Governance and conflicts of interest.
5. Disclosure and reporting.
6. Market integrity.
7. Cross-border cooperation.
8. Financial-stability monitoring.

12 Law, institutions, and monetary authority

12.1 Legal tender and monetary use

Legal tender status, contractual acceptability, settlement practice, and actual monetary adoption are distinct concepts.

A jurisdiction may permit Bitcoin transactions without granting Bitcoin legal-tender status.

Conversely, a legal-tender declaration does not guarantee substantial adoption.

The El Salvador experience illustrates the importance of distinguishing statutory status from observed usage.

The IMF's assessments of El Salvador discuss adoption, public-sector exposure, financial inclusion, fiscal risk, and the legal and regulatory implications of Bitcoin policy.

The case should be treated as an institutional example, not as a universal estimate of the effects of Bitcoin adoption.

12.2 Constitutional and statutory dimensions

A complete country-level analysis should examine:

1. Constitutional monetary provisions.
2. Central-bank enabling legislation.
3. Currency and legal-tender statutes.
4. Payment-system legislation.
5. Banking and prudential regulation.
6. Foreign-exchange controls.
7. Tax treatment of digital assets.
8. Property and contract law.
9. Insolvency and custody rules.
10. Anti-money-laundering obligations.

The legal analysis must be jurisdiction-specific.

There is no universal legal consequence that follows automatically from an increase in Bitcoin transaction share.

12.3 Institutional independence

Central-bank independence can be decomposed into:

$$I^G = \text{goal independence}, \tag{118}$$
$$I^T = \text{target independence}, \tag{119}$$
$$I^O = \text{operational independence}, \tag{120}$$
$$I^F = \text{financial independence}. \tag{121}$$

The effective autonomy of a central bank depends on the interaction of these institutional dimensions with the economic environment.

13 Physics, chemistry, biology, and information theory

13.1 Physics: networks and stability

The network representation of financial dependencies creates a useful analogy with dynamical systems.

A monetary system can be represented by a state vector $\mathbf{x}_t$.

Its evolution may be written as

$$\mathbf{x}_{t+1} = F(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\eta}_t). \tag{122}$$

Here:

$$\mathbf{u}_t = \text{policy interventions}, \tag{123}$$

$$\boldsymbol{\eta}_t = \text{external shocks}. \tag{124}$$

The objective is to identify whether policy interventions remain effective under changing network conditions.

The analogy with physical systems is methodological: stability, feedback, propagation, and resilience are common analytical concepts.

It does not imply that an economy obeys the same laws as a physical material.

13.2 Chemistry: composition and concentration

A financial system can be analysed as a composition of different monetary and settlement instruments.

Let the payment composition vector be

$$\mathbf{p}_t = \begin{bmatrix} p_t^D \\ p_t^B \\ p_t^S \\ p_t^O \end{bmatrix}, \tag{125}$$

where the components represent domestic currency, Bitcoin, stablecoins, and other instruments.

Subject to consistent measurement,

$$\sum_j p_t^j = 1. \tag{126}$$

This compositional structure is important because an increase in one share necessarily changes the relative shares of other components.

Compositional-data methods may therefore be more appropriate than separate unconstrained regressions on every payment share.

13.3 Biology: adaptation and resilience

Financial institutions adapt to technological change.

They may develop:

1. New custody arrangements.
2. Alternative settlement channels.
3. Risk-management procedures.
4. New capital and liquidity policies.
5. Technological interoperability.

The analogy with biological adaptation concerns the capacity of institutions to respond to environmental change.

It should not be interpreted as literal biological evolution.

13.4 Information theory

Let X denote the monetary-policy shock and Y the resulting economic outcome.

The information content of X about Y can be represented by mutual information:

$$I(X;Y) = \sum_{x,y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right). \tag{127}$$

A monetary transmission system may be studied in terms of the extent to which policy signals predict subsequent economic outcomes.

However, mutual information does not establish causality.

It may be useful as a supplementary diagnostic for changing policy-outcome dependence.

14 Welfare economics

14.1 Social welfare

The welfare implications of monetary adoption cannot be inferred from transmission capacity alone.

Let social welfare be

$$W = \sum_{i=1}^{N} \omega_i U_i, \tag{128}$$

where:

$$U_i = \text{utility of individual } i, \tag{129}$$

$$\omega_i = \text{social welfare weight}. \tag{130}$$

A policy may improve payment access while increasing financial volatility.

It may reduce transaction costs while increasing consumer exposure to custody or market risks.

The net welfare effect depends on the distribution of costs and benefits.

14.2 Financial inclusion

Relevant indicators include:

1. Access to transaction accounts.
2. Cost of domestic payments.
3. Cost of remittances.
4. Access to savings instruments.
5. Access to credit.
6. Consumer protection.
7. Digital literacy.
8. Reliability of payment services.

Bitcoin adoption should not be treated as synonymous with financial inclusion. Financial inclusion is an outcome to be measured.

14.3 Distributional consequences

Let the welfare effect for household group g be

$$\Delta W_g = \Delta U_g^{payments} + \Delta U_g^{access} - \Delta C_g^{volatility} - \Delta C_g^{risk}. \tag{131}$$

The aggregate effect may conceal unequal outcomes.

A complete analysis should therefore estimate heterogeneous effects across income, geography, financial access, and economic activity.

15 Political economy, geopolitics, and diplomacy

15.1 Monetary networks as institutional infrastructure

International monetary systems depend on networks of:

1. Central banks.
2. Commercial banks.
3. Payment providers.
4. Clearing and settlement systems.
5. Reserve institutions.
6. International financial organisations.
7. Regulatory authorities.

Alternative settlement networks may affect the geography of financial intermediation.

The extent of that effect depends on adoption, liquidity, legal recognition, convertibility, and network integration.

15.2 Geopolitical implications

The international monetary system is influenced by reserve-currency arrangements, payment infrastructure, capital flows, sanctions, and geopolitical alliances.

Bitcoin may provide an alternative transfer mechanism under certain conditions.

However, it does not automatically replace reserve currencies or the institutional functions of central banks.

The geopolitical analysis must distinguish:

$$G_1 = \text{payment-system diversification}, \tag{132}$$
$$G_2 = \text{reserve diversification}, \tag{133}$$
$$G_3 = \text{settlement independence}, \tag{134}$$
$$G_4 = \text{financial sanctions exposure}, \tag{135}$$
$$G_5 = \text{currency internationalisation}. \tag{136}$$

These are separate dimensions.

15.3 International cooperation

Cross-border digital-asset activity creates coordination problems.

A country acting alone may have limited ability to supervise foreign service providers, trace cross-border exposures, or address jurisdictional arbitrage.

Cooperation should include:

1. Regulatory information exchange.
2. Common reporting standards.
3. Cross-border supervisory coordination.
4. Cybersecurity incident cooperation.
5. Crisis-management protocols.
6. Consistent treatment of custodial assets.

The FSB framework provides an international baseline for considering such cooperation.

16 Warfare economics and strategic resilience

16.1 Financial infrastructure under stress

A sovereign monetary system may face disruptions from:

1. Cyberattacks.
2. Payment-system outages.
3. Sanctions.

4. Banking crises.

5. Cross-border liquidity interruptions.

6. Physical infrastructure failures.

7. Geopolitical conflict.

The relevant policy objective is resilience, not merely maximising the number of available payment technologies.

16.2 Continuity of essential monetary functions

Let the vector of essential monetary functions be

$$\mathbf{m}_t = \begin{bmatrix} m_t^{pay} \\ m_t^{settle} \\ m_t^{liq} \\ m_t^{credit} \\ m_t^{policy} \end{bmatrix}. \tag{137}$$

A resilient system should maintain minimum operational levels under stress:

$$m_{j,t} \geq \underline{m}_j \tag{138}$$

for essential functions j.

Redundancy can reduce single points of failure.

However, additional networks can introduce new dependencies, attack surfaces, and operational complexity.

16.3 Strategic conclusion

A sovereign state should not evaluate alternative monetary networks solely by their transaction speed or market value.

It should also evaluate their contribution to:

1. Continuity of payments.

2. Domestic liquidity management.

3. Financial stability.

4. Legal enforceability.

5. Crisis coordination.

6. Cyber resilience.

7. Monetary-policy effectiveness.

17 Cybersecurity and operational risk

17.1 Threat model

Relevant threats include:

1. Private-key compromise.
2. Exchange and custodian compromise.
3. Smart-contract vulnerabilities where applicable.
4. Ransomware and extortion.
5. Phishing and social engineering.
6. Supply-chain attacks.
7. Denial-of-service attacks.
8. Insider threats.
9. Data integrity failures.

17.2 Operational resilience

A useful risk representation is

$$Risk = Probability(Event) \times Impact(Event). \tag{139}$$

For interconnected systems, this simple expression is insufficient because risks may be correlated and cascading.

A more complete model should incorporate:

$$R^O = \text{operational risk}, \tag{140}$$
$$R^C = \text{cyber risk}, \tag{141}$$
$$R^L = \text{liquidity risk}, \tag{142}$$
$$R^M = \text{market risk}, \tag{143}$$
$$R^K = \text{counterparty risk}, \tag{144}$$
$$R^S = \text{systemic risk}. \tag{145}$$

A sovereign monetary architecture should evaluate the combined risk rather than each component in isolation.

18 Financial planning and sovereign risk

18.1 Public-sector exposure

Let the public sector hold Q_t units of Bitcoin.

Let its domestic-currency price be P_t.

The marked-to-market position is

$$V_t^{BTC} = Q_t P_t. \tag{146}$$

Its change is

$$\begin{aligned} \Delta V_t^{BTC} = Q_{t-1}\Delta P_t + P_{t-1}\Delta Q_t \\ + \Delta Q_t \Delta P_t. \end{aligned} \tag{147}$$

This decomposition separates price effects from quantity changes and their interaction.

A public-sector exposure analysis must also account for custody, liabilities, financing costs, and contingent obligations.

18.2 Stress testing

A scenario-based stress test may evaluate:

1. Bitcoin price decline.
2. Exchange liquidity contraction.
3. Custodian failure.
4. Domestic-currency depreciation.
5. Increased deposit withdrawals.
6. External reserve pressure.
7. Payment-system interruption.

Let the stress-loss function be

$$L_s = f(\Delta P_s^{BTC}, \Delta e_s, \Delta D_s, \Delta R_s, \Delta C_s). \tag{148}$$

The model should report losses and liquidity shortfalls under each scenario.

18.3 Budgetary discipline

Public investment or infrastructure expenditure should be evaluated through:

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}, \tag{149}$$

$$IRR = \text{discount rate at which } NPV = 0. \tag{150}$$

For public projects, a private-sector discount-rate calculation is not sufficient.

Social benefits, fiscal risk, contingent liabilities, distributional effects, and opportunity costs must also be considered.

19 Statistical architecture

19.1 Measurement hierarchy

The empirical architecture should separate:

Table 2: Proposed measurement architecture

Construct	Candidate measure	Principal concern
Bitcoin monetary use	Transaction share	Classification
Policy shock	Identified surprise	Endogeneity
Transmission	Outcome response	Channel selection
SDR exposure	Holdings and allocations	Interpretation
Monetary autonomy	Composite index	Weight sensitivity
Financial stability	Prudential indicators	Comparability
Institutional quality	Governance measures	Measurement error

19.2 Missing data

Missingness may be informative.

Let R_{it} indicate whether an observation is available.

The missing-data mechanism may be:

$$MCAR : R \perp X, Y, \tag{151}$$

$$MAR : R \perp Y \mid X, \tag{152}$$

$$MNAR : R \text{ depends on unobserved values.} \tag{153}$$

These mechanisms imply different statistical treatments.

A missing Bitcoin-use measure in a country with limited digital infrastructure should not automatically be interpreted as zero adoption.

19.3 Robustness

The empirical programme should include:

1. Alternative Bitcoin-use definitions.
2. Alternative SDR-exposure measures.
3. Alternative transmission outcomes.
4. Alternative fixed-effects structures.
5. Lag structures.
6. Outlier sensitivity.
7. Country exclusion tests.
8. Alternative sample windows.

9. Placebo outcomes.

10. Pre-trend diagnostics where applicable.

19.4 Multiple testing

If many outcomes and specifications are examined, the probability of false-positive findings increases.

The research should therefore distinguish primary hypotheses from exploratory analysis and consider appropriate multiple-testing adjustments.

20 Machine learning and computational analysis

20.1 Regime classification

The Sovereign Transmission Frontier may be complemented by a regime classifier.

Let the regime variable be

$$Z_{it} \in \{\text{Adequate}, \text{Near Frontier}, \text{Below Frontier}\}. \tag{154}$$

A classifier estimates

$$P(Z_{it} = k \mid \mathbf{X}_{it}). \tag{155}$$

Possible algorithms include:

1. Logistic regression.

2. Random forests.

3. Gradient-boosted trees.

4. Support-vector machines.

5. Hidden Markov models.

6. Bayesian regime-switching models.

The classifier should not replace the econometric identification strategy.

Prediction and causal inference answer different questions.

20.2 Regime-switching model

A latent regime model can be written as

$$T_{it} = \mu_{Z_{it}} + \boldsymbol{\theta}'_{Z_{it}} \mathbf{X}_{it} + \varepsilon_{it}. \tag{156}$$

The transition probability is

$$P(Z_{it} = j \mid Z_{i,t-1} = k) = p_{kj}. \tag{157}$$

This allows the transmission relationship to differ between institutional regimes.

20.3 Model validation

Validation should include:

1. Time-ordered train-test splits.
2. Out-of-sample evaluation.
3. Calibration analysis.
4. Class imbalance diagnostics.
5. Sensitivity to feature selection.
6. Stability across countries.
7. Drift detection.

Randomly splitting a time series into training and testing observations can create leakage.

20.4 Explainability

Feature importance does not establish causality.

A machine-learning model may identify variables that predict transmission shortfalls.

The econometric model must separately investigate whether those variables cause changes in transmission.

21 Algorithmic implementation

21.1 Threshold calculation

The following pseudocode computes the illustrative critical adoption share.

```
INPUT:
    initial_capacity = 0.80
    final_capacity   = 0.12
    minimum_capacity = 0.30

COMPUTE:
    slope = initial_capacity - final_capacity

    IF slope <= 0:
        REPORT that the assumed declining linear
        frontier is not valid.

    critical_share =
        (initial_capacity - minimum_capacity) / slope

CHECK:
    IF critical_share < 0 OR critical_share > 1:
```

```
        REPORT that no interior frontier exists.

OUTPUT:
    critical_share
    transmission_function
    sensitivity_table
```

21.2 Empirical estimation pipeline

```
PROCEDURE SOVEREIGN_TRANSMISSION_ANALYSIS:

    1. Define the country and authority sample.

    2. Collect monetary-policy shock data.

    3. Collect Bitcoin monetary-use data.

    4. Collect SDR holdings and allocations.

    5. Collect macroeconomic control variables.

    6. Validate units, dates, sources and definitions.

    7. Construct alternative adoption measures.

    8. Construct alternative SDR exposure measures.

    9. Estimate baseline fixed-effects models.

   10. Estimate interaction specifications.

   11. Test alternative lag structures.

   12. Examine institutional clustering.

   13. Conduct robustness and placebo tests.

   14. Estimate conditional marginal effects.

   15. Construct transmission-frontier estimates.

   16. Calculate uncertainty intervals.

   17. Perform out-of-sample validation.

   18. Report limitations and reproducibility metadata.

END PROCEDURE
```

21.3 Reproducibility

Each empirical result should be accompanied by:

1. Data-source metadata.
2. Data-access date.
3. Variable definitions.
4. Transformation procedures.
5. Software versions.
6. Random seeds where relevant.
7. Model specifications.
8. Estimation logs.
9. Test results.
10. Reproducible output files.

22 A proposed research architecture

The project can be organised into the following modules.

Table 3: Research modules

Module	Research object	Principal output
I	Monetary transmission	Transmission measures
II	Bitcoin monetary use	Adoption measures
III	SDR exposure	External exposure measures
IV	Econometric identification	Conditional estimates
V	Structural frontier	Critical thresholds
VI	Financial stability	Stress tests
VII	Legal architecture	Jurisdictional comparison
VIII	Geopolitics	International analysis
IX	Machine learning	Regime classification
X	Policy design	Resilience framework

22.1 Conceptual dependency graph

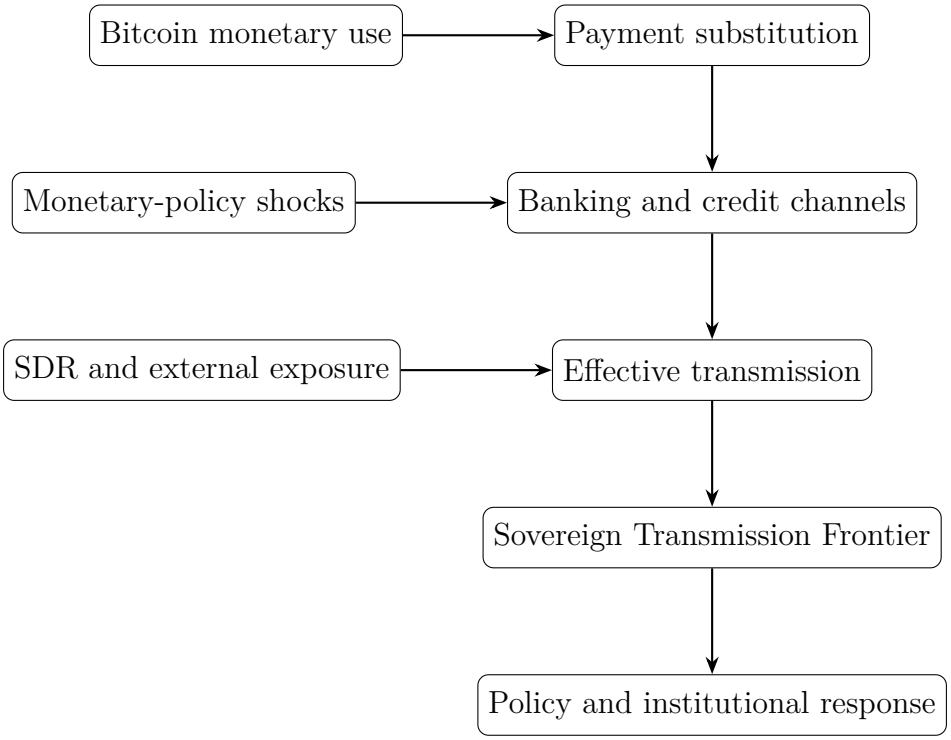


Figure 2: Conceptual structure of the Sovereign Transmission Frontier research programme.

23 Limitations and falsifiability

23.1 The illustrative graph is not an empirical estimate

The supplied graph establishes the following mathematical configuration:

$$T(b) = 0.80 - 0.68b. \tag{158}$$

It does not establish that the function is estimated from observed country-level data.

It does not establish that Bitcoin adoption causes a decline in monetary transmission.

It does not establish that 73.53 percent is a universal or optimal policy threshold.

The threshold is conditional on the selected intercept, slope, and minimum requirement.

23.2 Falsifiable propositions

The research programme should permit the following possibilities:

1. Bitcoin monetary use has no statistically detectable effect on transmission.
2. Bitcoin monetary use improves transmission through lower payment costs or better settlement access.
3. Bitcoin monetary use weakens transmission through deposit substitution or institutional fragmentation.

4. The effect varies with SDR exposure.
5. The effect varies with exchange-rate regimes.
6. The effect varies with financial development.
7. The relationship is nonlinear.
8. The relationship changes after regulatory reforms.

A credible theory must accommodate empirical rejection.

23.3 Identification limits

A statistically significant interaction does not by itself establish a structural causal mechanism.

The empirical analysis must establish:

1. Construct validity.
2. Measurement validity.
3. Identification validity.
4. Statistical reliability.
5. External validity.
6. Institutional interpretability.

24 Policy implications

24.1 Central-bank policy

Central banks should monitor the composition of payment systems and the channels through which monetary policy is transmitted.

The relevant objective is not necessarily to prevent the use of alternative payment technologies.

It is to preserve the capacity to implement monetary policy while maintaining financial stability and consumer protection.

24.2 Prudential policy

Prudential authorities should monitor material exposures among banks, payment institutions, custodians, and other systemically relevant intermediaries.

The monitoring framework should distinguish direct exposure from indirect exposure.

24.3 International monetary cooperation

International organisations and national authorities should develop compatible approaches to data collection, supervision, and crisis management.

SDR-related measures should be incorporated carefully, with clear distinctions between allocations, holdings, reserve adequacy, and external obligations.

24.4 Public finance

Governments should disclose material public-sector digital-asset exposures and contingent liabilities.

Public investment decisions should be evaluated through transparent fiscal-risk and cost-benefit analysis.

24.5 Institutional resilience

The policy objective is to maintain essential monetary functions under changing technological and geopolitical conditions.

A resilient monetary architecture should be:

1. Legally coherent.
2. Operationally robust.
3. Financially stable.
4. Cybersecure.
5. Internationally interoperable.
6. Economically measurable.
7. Democratically accountable.

25 Conclusion

The Sovereign Transmission Frontier provides a framework for examining the operational limits of monetary policy in economies experiencing changes in their payment and settlement structures.

The central analytical distinction is between the legal authority to conduct monetary policy and the effective capacity to transmit monetary-policy decisions through the economy.

The illustrative linear function

$$T(b) = 0.80 - 0.68b \tag{159}$$

combined with a minimum transmission requirement of 0.30, yields

$$b^* = \frac{25}{34} \approx 0.735294. \tag{160}$$

This establishes an internally consistent mathematical frontier for the supplied graphical configuration.

The next stage is empirical identification.

The integrated specification introduces monetary-policy shocks, Bitcoin monetary use, SDR exposure, and their interactions. It provides a way to test whether the transmission of monetary policy varies with the composition of domestic payment systems and external monetary constraints.

The analysis also demonstrates why Bitcoin ownership, transaction activity, legal-tender status, reserve exposure, and monetary substitution must be measured separately.

The institutional implications extend beyond central banking. They concern fiscal sustainability, financial stability, payment infrastructure, cybersecurity, international law, regulatory cooperation, welfare, and geopolitical resilience.

The frontier is therefore not a prediction of monetary collapse.

It is a proposed analytical boundary for identifying when a monetary authority's effective transmission capacity may become inadequate relative to a specified policy requirement.

The ultimate objective is to construct a measurable, falsifiable, and institutionally useful theory of sovereign monetary transmission.

$$\boxed{\text{Monetary Sovereignty} = \text{Authority} + \text{Capacity} + \text{Transmission} + \text{Resilience}} \tag{161}$$

A sovereign monetary system is not fully characterised by the instruments it legally controls.

It must also be understood through the economic mechanisms, institutions, networks, and infrastructures that allow its decisions to have effect.

The Sovereign Transmission Frontier

A Mathematical appendix

A.1 General linear frontier

For

$$T(b) = \alpha - \beta b, \tag{162}$$

the frontier is

$$b^* = \frac{\alpha - \underline{T}}{\beta}. \tag{163}$$

The existence condition is

$$\alpha - \beta \leq \underline{T} \leq \alpha, \tag{164}$$

assuming $\beta > 0$.

A.2 Threshold uncertainty

If the estimated parameters are uncertain, the estimated threshold is

$$\widehat{b}^* = \frac{\widehat{\alpha} - \underline{T}}{\widehat{\beta}}. \tag{165}$$

A first-order delta-method approximation gives

$$\mathrm{Var}(\widehat{b}^*) \approx \nabla g' \widehat{\Sigma} \nabla g, \tag{166}$$

where

$$g(\alpha, \beta) = \frac{\alpha - \underline{T}}{\beta}. \tag{167}$$

The gradient is

$$\nabla g = \begin{bmatrix} \frac{1}{\beta} \\ -\frac{\alpha - \underline{T}}{\beta^2} \end{bmatrix}. \tag{168}$$

Here $\widehat{\Sigma}$ is the estimated covariance matrix of $(\widehat{\alpha}, \widehat{\beta})$.

A bootstrap procedure may be preferable when the threshold is near a boundary or the estimator is nonlinear.

A.3 Threshold confidence sets

A confidence interval for the threshold should account for uncertainty in both the intercept and slope.

The uncertainty in the minimum requirement should also be incorporated if it is estimated rather than fixed.

A policy threshold should therefore be reported with its assumptions and uncertainty, not as a single unqualified point estimate.

References

[1] Akerlof, G. A. (1970). The Market for "Lemons": Quality Uncertainty and the Market Mechanism. *Quarterly Journal of Economics*, 84(3), 488–500.

[2] Angrist, J. D., and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

[3] Auer, R., and Böhme, R. (2020). The Technology of Retail Central Bank Digital Currency. BIS Quarterly Review.

[4] Bank for International Settlements (2021). *Annual Economic Report 2021*. BIS, Basel.

[5] Borio, C. (2014). The Financial Cycle and Macroeconomics: What Have We Learnt? *Journal of Banking and Finance*, 45, 182–198.

[6] Brunnermeier, M. K., and Sannikov, Y. (2014). A Macroeconomic Model with a Financial Sector. *American Economic Review*, 104(2), 379–421.

[7] Cochrane, J. H. (2005). *Asset Pricing*. Revised Edition. Princeton University Press.

[8] Diamond, D. W., and Dybvig, P. H. (1983). Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy*, 91(3), 401–419.

[9] Engle, R. F., and Granger, C. W. J. (1987). Co-integration and Error Correction: Representation, Estimation, and Testing. *Econometrica*, 55(2), 251–276.

[10] Financial Stability Board (2023). *High-Level Recommendations for the Regulation, Supervision and Oversight of Crypto-Asset Activities and Markets*. Financial Stability Board.

[11] Friedman, M. (1968). The Role of Monetary Policy. *American Economic Review*, 58(1), 1–17.

[12] Gertler, M., and Karadi, P. (2015). Monetary Policy Surprises, Credit Costs, and Economic Activity. *American Economic Journal: Macroeconomics*, 7(1), 44–76.

[13] Goodhart, C. A. E. (1998). The Two Concepts of Money: Implications for the Analysis of Optimal Currency Areas. *European Journal of Political Economy*, 14(3), 407–432.

[14] Hayek, F. A. (1976). *Denationalisation of Money*. Institute of Economic Affairs.

[15] International Monetary Fund. *Articles of Agreement of the International Monetary Fund*. IMF, Washington, D.C.

[16] International Monetary Fund. *Special Drawing Rights (SDR)*. IMF official SDR documentation and statistical database.

[17] International Monetary Fund (2026). *SDR Allocations and Holdings for All Members*. September 30, 2026.

[18] International Monetary Fund (2025). *El Salvador: Article IV Consultation and Selected Issues Reports*. IMF Country Reports.

[19] Kiyotaki, N., and Moore, J. (1997). Credit Cycles. *Journal of Political Economy*, 105(2), 211–248.

[20] Lucas, R. E. (1976). Econometric Policy Evaluation: A Critique. *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46.

[21] Mankiw, N. G. (1991). The Reincarnation of Keynesian Economics. *European Economic Review*, 35(1), 165–171.

[22] Merton, R. C. (1974). On the Pricing of Corporate Debt: The Risk Structure of Interest Rates. *Journal of Finance*, 29(2), 449–470.

[23] Mishkin, F. S. (1996). The Channels of Monetary Transmission: Lessons for Monetary Policy. NBER Working Paper No. 5464.

[24] Mundell, R. A. (1961). A Theory of Optimum Currency Areas. *American Economic Review*, 51(4), 657–665.

[25] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.

[26] North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.

[27] Obstfeld, M., and Rogoff, K. (1996). *Foundations of International Macroeconomics*. MIT Press.

[28] Rogoff, K. (2016). *The Curse of Cash*. Princeton University Press.

[29] Sargent, T. J. (1987). *Macroeconomic Theory*. Second Edition. Academic Press.

[30] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27, 379–423 and 623–656.

[31] Sims, C. A. (1980). Macroeconomics and Reality. *Econometrica*, 48(1), 1–48.

[32] Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press.

Glossary

Bitcoin monetary adoption The extent to which Bitcoin is used in monetary transactions, payments, or settlement.

Bitcoin transaction share The proportion of a defined transaction aggregate attributable to Bitcoin-based payments or settlement.

Central-bank autonomy The effective institutional and operational capacity of a monetary authority to conduct monetary policy.

Critical adoption share The adoption level at which the modelled transmission capacity equals a specified minimum requirement.

Effective monetary transmission The degree to which monetary-policy actions influence economic outcomes through relevant transmission channels.

Endogeneity A condition in which an explanatory variable is correlated with the regression error term.

External monetary exposure A country's exposure to international reserve, financing, exchange-rate, and external liquidity conditions.

Monetary-policy shock An unexpected component of monetary policy that is isolated from systematic policy responses.

Monetary sovereignty The legal, institutional, and operational capacity of a state to organise and conduct monetary affairs.

SDR Special Drawing Right, an international reserve asset created by the International Monetary Fund.

SDR allocation An allocation of SDRs to an eligible IMF member under the applicable institutional framework.

SDR holdings The SDR balance held by an IMF member.

Sovereign Transmission Frontier The boundary at which effective monetary transmission capacity equals a specified minimum operational requirement.

Structural identification The process of identifying causal economic relationships using explicit assumptions and restrictions.

Transmission coefficient A parameter measuring the relationship between a monetary-policy instrument or shock and an economic outcome.

The End